



KOLEJ MATRIKULASI MELAKA
UNIT MATEMATIK
SEMESTER I, SESI 2024/2025
MATHEMATICS 1 SM015
KARISMATIK E



ANSWER ALL QUESTIONS

SECTION A [25 marks]

BAHAGIAN A [25 markah]

This section consists of 7 questions. Answer all questions.
Bahagian ini mengandungi 7 soalan. Jawab semua soalan.

1. Given that $f(x) = \frac{(x-4)^2}{x^2}$
 - (a) Find $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 0^-} f(x)$. Deduce the vertical asymptote of $f(x)$.
(4 marks)
 - (b) Find $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$. Deduce the horizontal asymptote of $f(x)$.
(4 marks)
2.
 - (a) Find $\frac{dy}{dx}$ for $y = x^2 \cos x^2$. (4 marks)
 - (b) Given $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$, then show that $\frac{dy}{dx} = \frac{4}{(e^x + e^{-x})^2}$. (4 marks)
3.
 - (a) Find the critical number of $f(x) = \sqrt{x^2 - 16}$. (5 marks)
 - (b) Find the interval(s) on which f is increasing and decreasing. (4 marks)





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SECTION B [75 marks]
BAHAGIAN B [75 markah]

This section consists of 7 questions. Answer all questions.
Bahagian ini mengandungi 7 soalan. Jawab semua soalan.

1. A complex number, z , is such that $|z - 1| = 2\sqrt{2}$ and $\arg(z - 1) = -\frac{1}{4}\pi$. Express z in the form $a + bi$, where a and b are real constants. (5 marks)

2. (a) Find the only real solution of the logarithmic equation
 $2\log_2 x + \log_2(x - 1) - \log_2(5x + 4) = 1$. (5 marks)

- (b) i) Given $|t| = 3$, find the possible values of $|2t - 1|$. (3 marks)

- ii) Solve the inequality $|x - \sqrt{2}| > |x + 3\sqrt{2}|$. (3 marks)

3. Given that f, g, h are as follows :
$$f(x) = |x|$$
$$g(x) = x^2 - 2$$
$$h(x) = \frac{1}{x}, \quad x \neq 0$$
 - i) Find $F(x) = f \circ g \circ h(x)$ (3 marks)
 - ii) Find the values of x when $F(x) = 5$ (3 marks)

4. (a) Given that $f : x \mapsto \ln(3x - 2) + 5$,
 - i) Find f^{-1} , and state its domain and range. (4 marks)
 - ii) Hence, show the relationship between the graphs of f and f^{-1} on a diagram. (3 marks)

- (b) The function g and h are defined by $g(x) = \frac{x+1}{x-2}$ and $h(x) = \frac{ax+3}{x}$. Given that $hg^{-1}(4) = 6$, calculate the value of a . (7 marks)





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5. (a) Find the vertical and horizontal asymptotes of the graph of $f(x) = \frac{4x}{x^3 + 8}$.
(6 marks)

(b) The function f is defined by

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2} & , \quad x < 2 \\ a^2 - 2 & , \quad x = 2 \\ \frac{|2 - x|}{x - 2} + b & , \quad x > 2 \end{cases}$$

where a and b are constants.

- i) If $\lim_{x \rightarrow 2} f(x)$ exists, find the value of b . (5 marks)
- ii) If $f(x)$ is continuous at $x = 2$, determine the value of a . (3 marks)
- 6) An inverted conical tank with radius 3 meter and height 4 meter is fully filled with **water**. The **water** is leaking from the vertex of the tank at a rate of $0.2 \text{ m}^3 \text{ s}^{-1}$. At the instant when the height of **water** is 2 meters, calculate the rate of change of
- (a) the water level. (7 marks)
- (b) the radius of the water surface. (2 marks)

7. (a) Given $2y^2 - 3xy + x^2 = 2$.
- i) By using implicit differentiation, show that
- $$(-3y + 2x) \frac{d^2x}{dy^2} + 2 \left(\frac{dx}{dy} \right)^2 - 6 \frac{dx}{dy} + 4 = 0 \quad (4 \text{ marks})$$
- ii) Find the values of $\frac{d^2x}{dy^2}$ when $x = 0$. (5 marks)





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(b) Given the parametric equations $x = 2 \cos^3 t$ and $y = 4 \sin^3 t$.

i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. (4 marks)

ii) Find the exact value of $\frac{d^2y}{dx^2}$ when $t = \frac{\pi}{3}$. (3 marks)

END OF QUESTION PAPER
KERTAS SOALAN TAMAT

